

Experiment - 3.1.1. Largest of Three Numbers

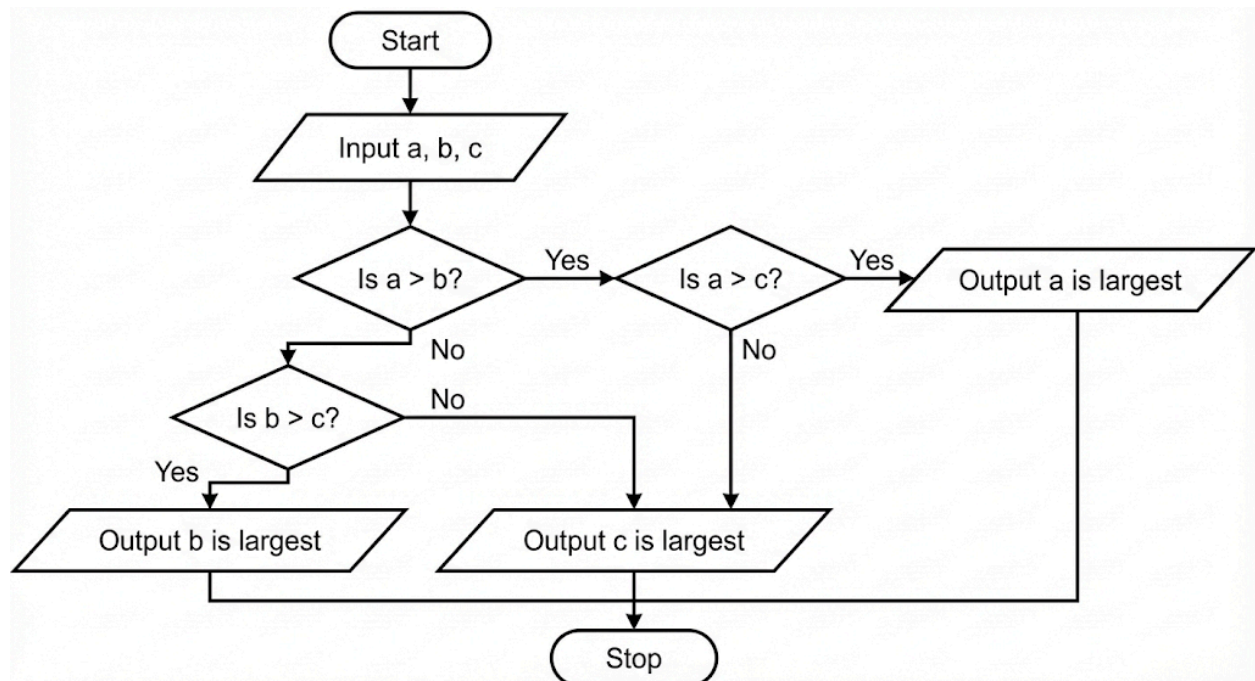
1. Aim

To design and implement a Python program that prompts the user to enter three integers and determines the largest among them using conditional statements (if-elif-else).

2. Pseudocode

1. **START**
2. **READ** three integers a, b, and c from the user (one per line).
3. **INITIALIZE** a variable largest to 0.
4. **IF** $a \geq b$ AND $a \geq c$:
 - Set largest = a.
5. **ELSE IF** $b \geq a$ AND $b \geq c$:
 - Set largest = b.
6. **ELSE**:
 - Set largest = c.
7. **PRINT** the value of largest.
8. **END**

3. Flowchart



4. Python Program

Program to find the largest of three integers

Input: Three integers, one per line
Output: The largest integer among the three

Taking three integers as input

```
a = int(input(""))
```

```
b = int(input(""))
```

```
c = int(input(""))
```

Initializing variable to store the largest value

```
largest = 0
```

Comparative logic using logical 'and'

```
if (a >= b) and (a >= c):
```

```
    largest = a
```

```
elif (b >= a) and (b >= c):
```

```
    largest = b
```

```
else:
```

```
    largest = c
```

Displaying the result

```
print(largest)
```

5. Experiment Screenshot

The screenshot shows the CODETANTRA IDE interface. On the left, the problem statement for "3.1.1. Largest of Three Numbers" is displayed, including input and output formats. The main editor shows a Python program that takes three integers as input and prints the largest one. The program is submitted, and the results show that it passed all test cases. The test case details for Test case 1 are also visible, showing the expected and actual outputs.

```
1 a = int(input(""))
2 b = int(input(""))
3 c = int(input(""))
4
5 largest = 0
6 if (a >= b) and (a >= c):
7     largest = a
8 elif (b >= a) and (b >= c):
9     largest = b
10 else:
11     largest = c
12
13 print(largest)
```

Test case 1 (12 ms)

Expected output

5

6

7

Actual output

5

Experiment - 3.1.2. Celsius to Fahrenheit

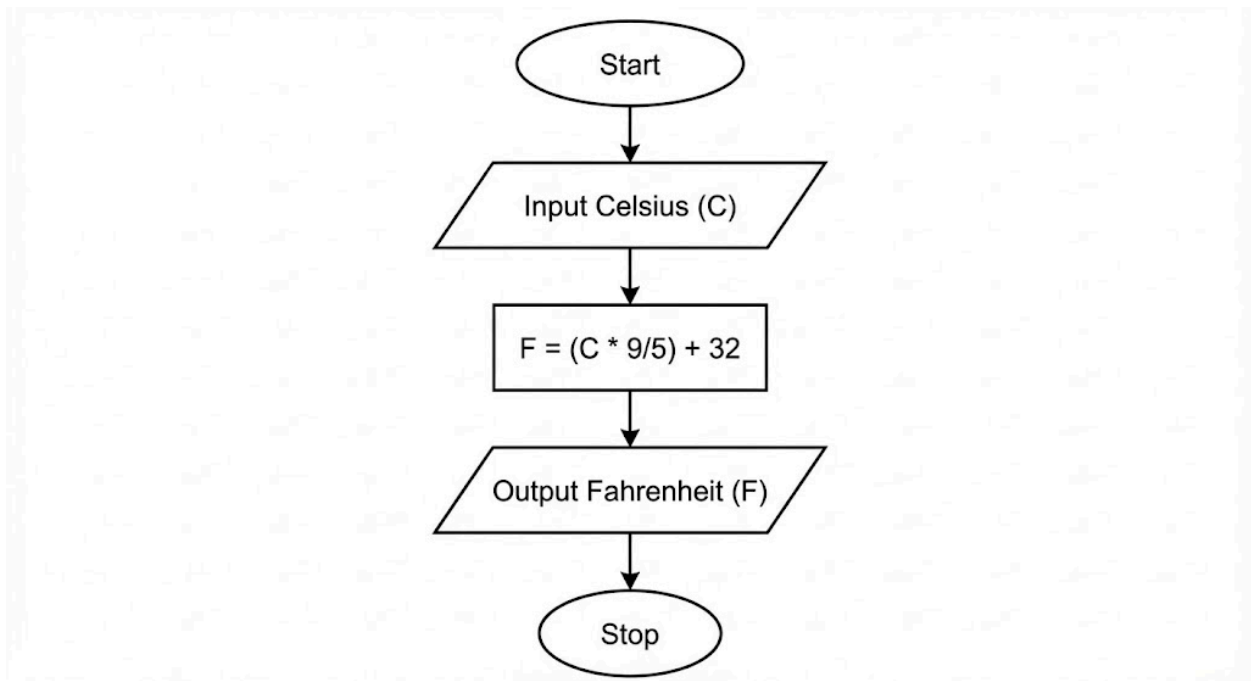
1. Aim

To design and implement a Python program that converts temperature from Celsius to Fahrenheit. The program accepts a floating-point value representing Celsius and applies the formula $Fahrenheit = (Celsius \times \frac{9}{5}) + 32$, displaying the result formatted to two decimal places.

2. Pseudocode

1. **START**
2. **READ** the input value as a float and store it in variable a (Celsius).
3. **CALCULATE** the Fahrenheit value using the formula: $b = a * (9/5) + 32$.
4. **STORE** the result in variable b.
5. **FORMAT** the value of b to 2 decimal places.
6. **PRINT** the formatted result.
7. **END**

3. Flowchart



4. Python Program

```
# Program to convert Celsius to Fahrenheit
# Input: Celsius value as a float
# Output: Fahrenheit value formatted to 2 decimal places
```

```
# Taking Celsius input from the user
a = float(input(""))
```

```
# Applying the conversion formula
b = a * (9/5) + 32
```

```
# Displaying the result formatted to 2 decimal places
print(f"{b:.2f}")
```

5. Experiment Screenshot

The screenshot displays the CODETANTRA IDE interface. On the left, the problem statement for "3.1.2. Celsius to Fahrenheit" is shown, including the formula $Fahrenheit = (Celsius \times \frac{9}{5}) + 32$ and input/output formats. The main editor pane shows the following Python code:

```
1 a = float(input(""))
2 b = a*(9/5) + 32
3 print(f"{b:.2f}")
4
```

Below the code, the Test cases pane indicates that all tests passed:

- Average time: 0.002 s (2.38 ms)
- Maximum time: 0.004 s (4.00 ms)
- 4 out of 4 shown test case(s) passed
- 4 out of 4 hidden test case(s) passed

Test case 1 details:

Expected output	Actual output
0.0	0.0
32.00	32.00

At the bottom, there are buttons for "Terminal", "Test cases", "Prev", "Reset", "Submit", and "Next".